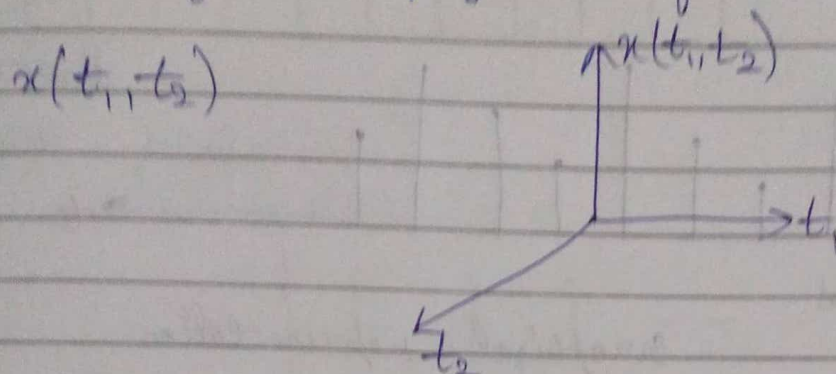


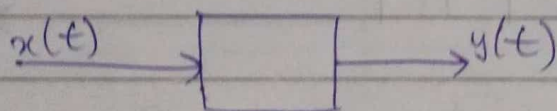
* Signal / input / output / function / waveform:

These are functions of one or more independent variables which contain information about the nature of some physical formula.



$y(t_1, t_2, t_3) \rightarrow$ graphical representation is not possible in 3D space.
 $z(t)$

* System: This is an interconnection of components / devices / processes with ^{access} ports through which energy (signal) can be applied as an input or extracted as an output.



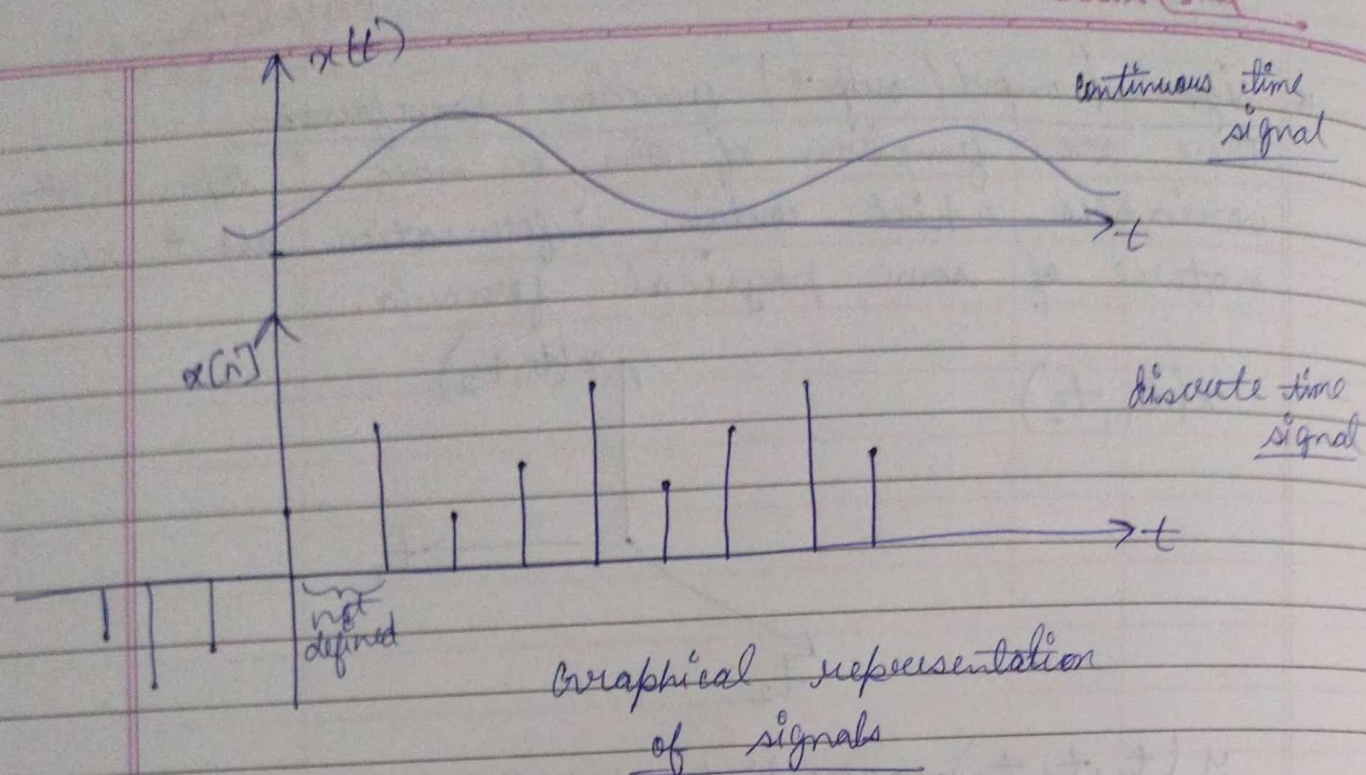
Classification of signals:

① Continuous vs discrete time signals

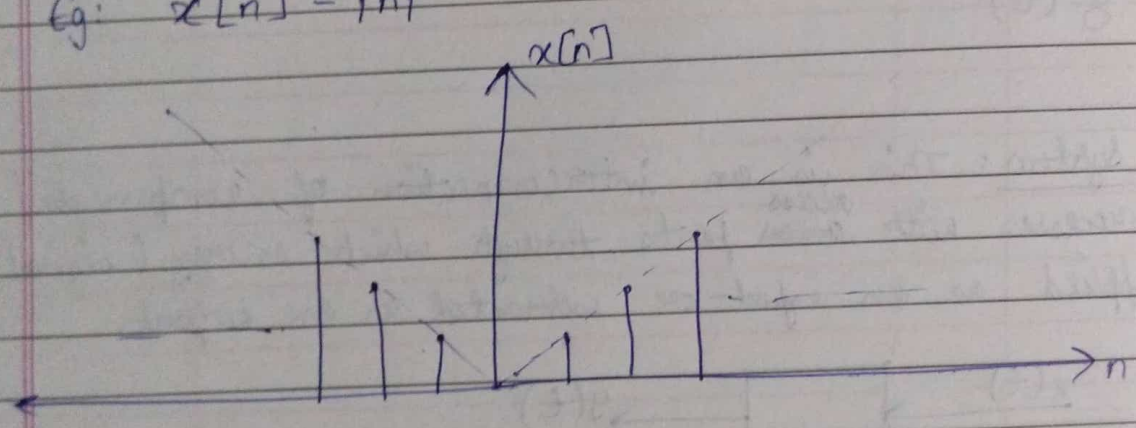
Representation

$x(t)$; $t \in \mathbb{R} \rightarrow$ continuous time signal
 Eg: $x(t) = \sin\{2\pi k f_0 t\}$, k is an integer.

$x[n]$; $n \in \mathbb{Z} \rightarrow$ discrete time signal
 Eg: $x[n] = x\left(\frac{n}{5f_0}\right) = \sin\left\{\frac{2\pi n k}{5}\right\} \leftarrow$ discrete signal
 derived from continuous signal



Eg: $x[n] = |n|$



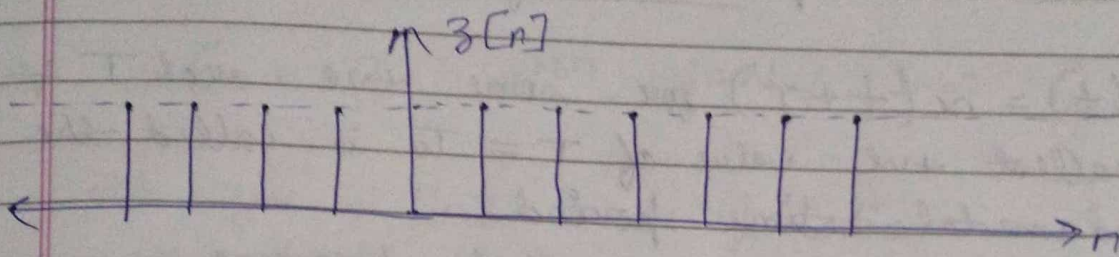
$x[n] = \{ \dots, 3, 2, 1, 0, 1, 2, 3, \dots \}$
 ↑ the arrow shows $(n=0)$ value.

∅ $x(n) = |n|$, then plot
 $y(n) = n$ $z[n] = \frac{x[x] + y[y]}{2}$

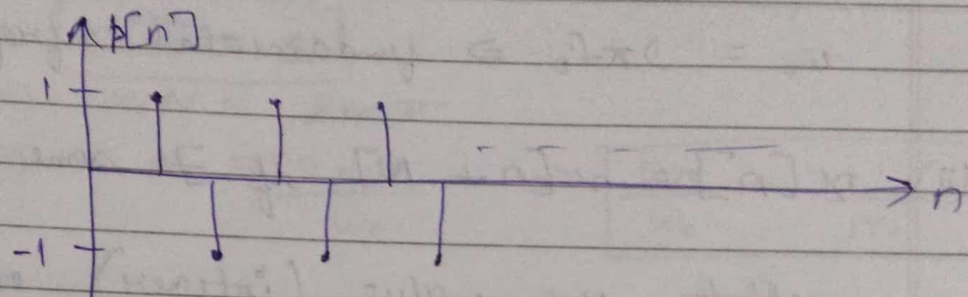
$$= \frac{|n| + n}{2}$$

$$\Rightarrow \begin{cases} n & , n \geq 0 \\ 0 & , n < 0 \end{cases}$$

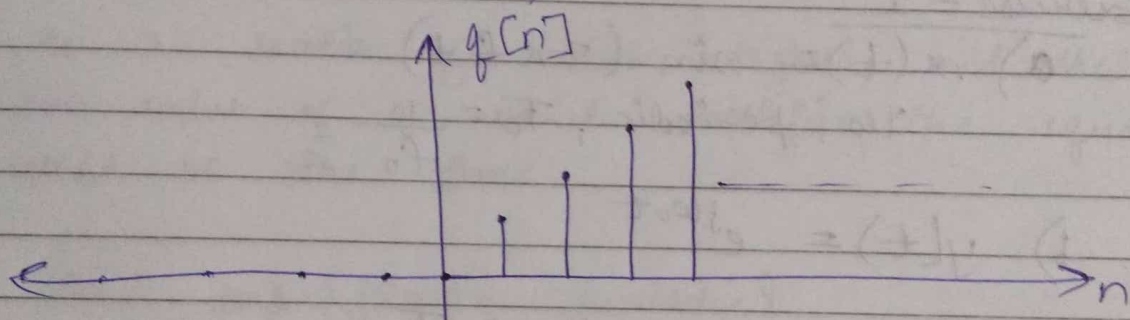
Eg: Plot $z[n] = 2$



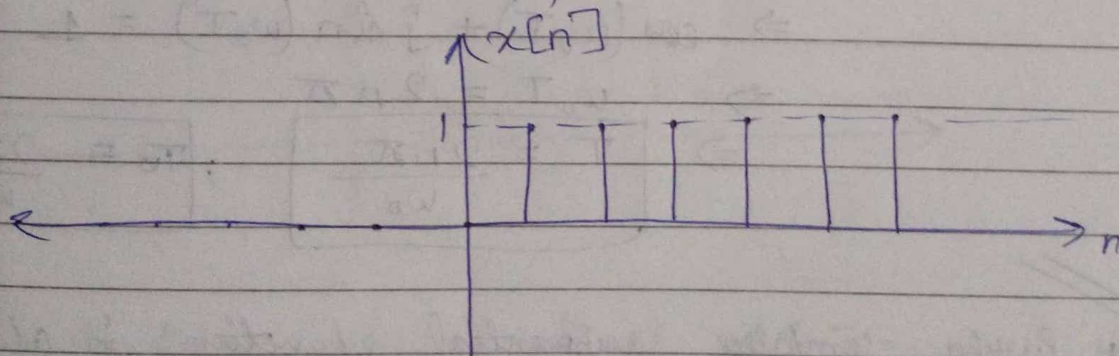
Eg: Plot $p[n] = (-1)^n$



Eg: Plot $q[n] = \begin{cases} n & ; n \geq 0 \\ 0 & ; n < 0 \end{cases}$



Eg: Plot $x[n] = \begin{cases} 1 & ; n \geq 0 \\ 0 & ; n < 0 \end{cases}$



② Periodic vs Aperiodic signal -

(i) $x(t) = x(t+T)$ for some +ve real $T \neq 0$
 smallest +ve value of $T = T_0$ is called the fundamental time period.

$$f_0 = \frac{1}{T_0} \Rightarrow \text{fundamental frequency}$$

$$\omega_0 = 2\pi f_0 \Rightarrow \text{fundamental frequency}$$

(ii) $x[n] = x[n+N]$ if $\exists$ some +ve $N \in \mathbb{Z} \neq 0$
 smallest +ve value (integer) of $N = N_0$ is

Exercise - 1

a) $x(t) = \sin(2\pi f_0 t)$

Yes, periodic, $T_0 = \frac{1}{f_0}$

b) $y(t) = e^{j\omega_0 t}$

$$y(t+T) = e^{j\omega_0 t} \cdot e^{j\omega_0 T} = y(t)$$

$$\Rightarrow e^{j\omega_0 T} = 1$$

$$\Rightarrow \cos(\omega_0 T) + j \sin(\omega_0 T) = 1$$

$$\Rightarrow \omega_0 T = 2n\pi$$

$$\Rightarrow \boxed{T = \frac{2n\pi}{\omega_0}}$$

$$\boxed{T_0 = \frac{2\pi}{\omega_0}}$$

* Purely complex exponential function is always periodic with a time period $T_0 = \frac{2\pi}{\omega_0}$

Ex-2) $x[n] = e^{j\omega_0 n}$

$x[n+N] = e^{j\omega_0 n} \cdot e^{j\omega_0 N} = e^{j\omega_0 n}$ (for periodicity)

$\Rightarrow$ for $x[n]$ to be periodic, we should find some +ve integer value of N such that

$$\boxed{e^{j\omega_0 N} = 1}$$

$\Rightarrow \omega_0 N = 2m\pi$

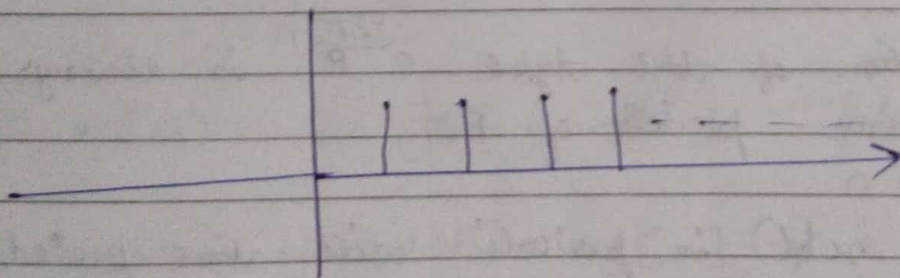
$\Rightarrow N = \frac{2m\pi}{\omega_0} \Rightarrow \boxed{\frac{2\pi}{\omega_0} = \frac{N}{m}}$

* A purely complex discrete time exponential function is periodic iff $\frac{\omega_0}{2\pi}$ can be expressed

as a ratio of two integers, and the smallest +ve value of N which satisfies the equation would be the time.

Ex-3) Is $\exp\{j2\pi n\}$ periodic?

Ans $\Rightarrow e^{j2\pi n} = (e^{j2\pi})^n = 1$



Ex-4) Consider the signal $x[n] = \exp\left\{j\frac{13\pi}{7}n\right\}$

Sol $e^{j\frac{13\pi}{7}N} = 1$

$\Rightarrow \frac{13\pi}{7}N = \cancel{2\pi k} 2\pi p \quad p = 1, 2, 3, \dots$

$\Rightarrow N = \frac{14\cancel{\pi}p}{13\cancel{\pi}} \Rightarrow \boxed{N = \frac{14p}{13}}, \quad \boxed{N_0 = 14}$

$\Rightarrow e^{\frac{j \cdot 13 \cdot 2\pi \cdot n}{14}} = e^{j\left(\frac{14 \cdot 2\pi \cdot n}{14} - \frac{2\pi \cdot n}{14}\right)}$
 $= e^{j2\pi n} \cdot e^{-\frac{j2\pi n}{14}}$
 $= e^{-\frac{j2\pi n}{14}}$

$\omega_0 = \frac{2\pi}{14} \Rightarrow \frac{\omega_0}{2\pi} = \frac{1}{14}$

$\Rightarrow e^{j\frac{2\pi}{P}n} \cdot e^{j\frac{2\pi}{P}N} = e^{j\frac{2\pi}{P}n}$

$\Rightarrow e^{j\frac{2\pi}{P}N} = 1$

$\Rightarrow \frac{2\pi N}{P} = 2\pi m$

$\Rightarrow \boxed{N = mP}$

$\boxed{N_0 = P}$

* A function of the type $e^{j\frac{2\pi}{P}n}$ is always periodic with a time period of P.

Ex-5) A signal $x(t)$ is periodic with time period T_x , $y(t)$ is periodic with time period T_y , & $z(t) = a x(t) + b y(t)$. Is $z(t)$ periodic?

Sol $\left. \begin{aligned} x(t) &= x(t + nT_x) \quad \forall n \in \mathbb{Z} \\ y(t) &= y(t + mT_y) \quad \forall m \in \mathbb{Z} \end{aligned} \right\} \textcircled{1}$

Does there exist some T such that
 $g(t) = g(t+T) \quad \forall t$

$$ax(t+T) + by(t+T) = ax(t) + by(t) \quad \text{for some } T$$

from ①, ② we get

$$T = nT_x \quad \text{and} \quad T = mT_y$$

Ex-6) Given that $x(t)$ is periodic with time period T , find out the time period of $x(2t)$ and $x(t) + x(2t)$ and $x(t) + x(2t) + x(t/2)$.

Sol $\rightarrow x(t) = x(t+nT) \quad n \in \mathbb{Z}$

$$y(t+T_1) = x(t+T_1) + x(2t+2T_1) + x\left(\frac{t}{2} + \frac{T_1}{2}\right)$$

$$= x(t) + x(2t) + x(t/2)$$

$$\left. \begin{array}{l} T_1 = nT \\ \text{and } T_1 = mT/2 \\ \text{and } T_1 = 2pT \end{array} \right\} \Rightarrow \boxed{T_1 = 2T}$$

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Ex-1) $y[n] = \cos[\pi k(n+1)^2]$ periodic or not?
 $n \in \mathbb{Z}; k$ is fixed.

Sol, Suppose, $y[n+P] = y[n] \quad \forall n$ & some integer P

$$\text{So, } y[n+P] = \cos[\pi k(n^2 + 2nP + P^2)]$$

$$= \cos[\pi k n^2 + 2\pi k n P + \pi k P^2] = \cos(\pi k n^2)$$

$$\Rightarrow \pi k(2nP + P^2) = 2\pi m$$

$$\Rightarrow k(2nP + P^2) = 2m$$

always an integer

always an integer even

* $K=1 \Rightarrow 2nP + P^2 \Rightarrow \text{even}$, so, P should be even $\Rightarrow \boxed{P=2}$

* $K=3 \Rightarrow 3(2nP + P^2) \Rightarrow \text{even}$, so, P should be even $\Rightarrow \boxed{P=2}$

* $K=2 \Rightarrow 2(2nP + P^2) \Rightarrow \text{even}$, so, P can be any +ve integer $\Rightarrow \boxed{P=1}$

$\Rightarrow$ so, the time period is 2 when k is odd & time period is 1 when k is even.

Ex-2) $y(t) = \cos(\pi k t^2)$ periodic or not?

Sol $y(t+T) = \cos(\pi k t^2 + 2\pi k t T + \pi k T^2) \quad \forall t$
 $= \cos(\pi k t^2)$

$\Rightarrow 2\pi k t T + \pi k T^2 = 2\pi p \quad p \in \mathbb{Z}$

$\Rightarrow 2k t T + k T^2 = 2p$

Hence, there is no value of T ^{necessarily} for which the quantity gives an integer.

So, this is not periodic.

Ex-3) a) Defining $y(t) = \sum_{k=-\infty}^{\infty} x(t+kP)$; $k \in \mathbb{Z}$; P is constant

Is $y(t)$ periodic? If yes, what is its time period?

Sol Let $y(t)$ be periodic for some value of T .

Sol $y(t+T) = \sum_{k=-\infty}^{\infty} x(t+T+kP) = y(t) \quad \forall t$

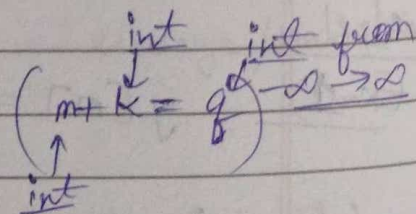
Let $T = mP$

$= \sum_{k=-\infty}^{\infty} x(t + (m+k)P)$

$= \sum_{q=-\infty}^{\infty} x(t + qP)$

$= y(t)$

$\Rightarrow$ time period = P



* Such a function ^(infinity summation) is always periodic with time period P .

b) $y[n] = \sum_{k=-\infty}^{\infty} x[n + kP]$ periodic or not?

Ex-1) If $x(t)$ & $y(t)$ are periodic, find out if $x(t) \cdot y(t)$ is periodic or not?

$x(t)/x[n]$ are

Ex-2) If $x(t)$ is periodic, then are these periodic or not?

a) $x^2(t)$

g) $x[n+5]$

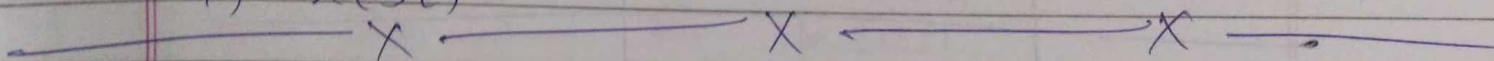
b) $x^2[n]$

c) $x(t^2)$

d) $x[n^2]$

e) $x(t+3)$

f) $x(3t)$



Ex-4) If $x[n]$ is periodic with time period = N , show that:

(i) $\sum_{k=n_0}^n x[k] = \sum_{k=n_0+N}^{n+N} x[k]$

(ii) $\sum_{k=0}^{N-1} x[k] = \sum_{k=n_0}^{n_0+N-1} x[k]$; $0 < n_0 \leq N-1$

Sol (i) $\sum_{k=n_0}^n x[k] = \sum_{k=n_0}^n x[k+N]$

define $k+N = p$

$= \sum_{p=n_0+N}^{n+N} x[p]$

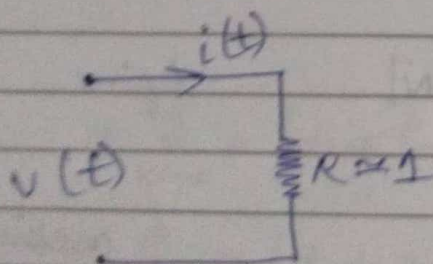
(ii) $\sum_{k=0}^{N-1} x[k] = \left(\sum_{k=0}^{n_0-1} x[k] \right) + \sum_{k=n_0}^{N-1} x[k] + \left(\sum_{k=N}^{N+n_0-1} x[k] \right)$

combine

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Energy vs Power signals:

	Continuous time	Discrete time
Periodic	Fourier series	
Aperiodic		



$$\begin{aligned}
 p(t) &= v(t) i(t) \\
 &= i^2(t) R \\
 &= \frac{v^2(t)}{R}
 \end{aligned}$$

$$p(t) = x^2(t) = |x(t)|^2$$

$$(1) E_x = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

① Energy signal is defined as a signal for which
 E_x is finite
 & P_x is zero.

② Power signal is defined as a signal for which
 P_x is finite
 & E_x is infinite.

Energy vs Power for DT signals:

$$E_x = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2$$

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Ex-1) $x(t) = e^{j\omega_0 t}$

Sol Recall: Periodic TP: $T_0 = \frac{2\pi}{\omega_0}$

$$E_x = \lim_{T \rightarrow \infty} \int |e^{j\omega_0 t}|^2 dt$$

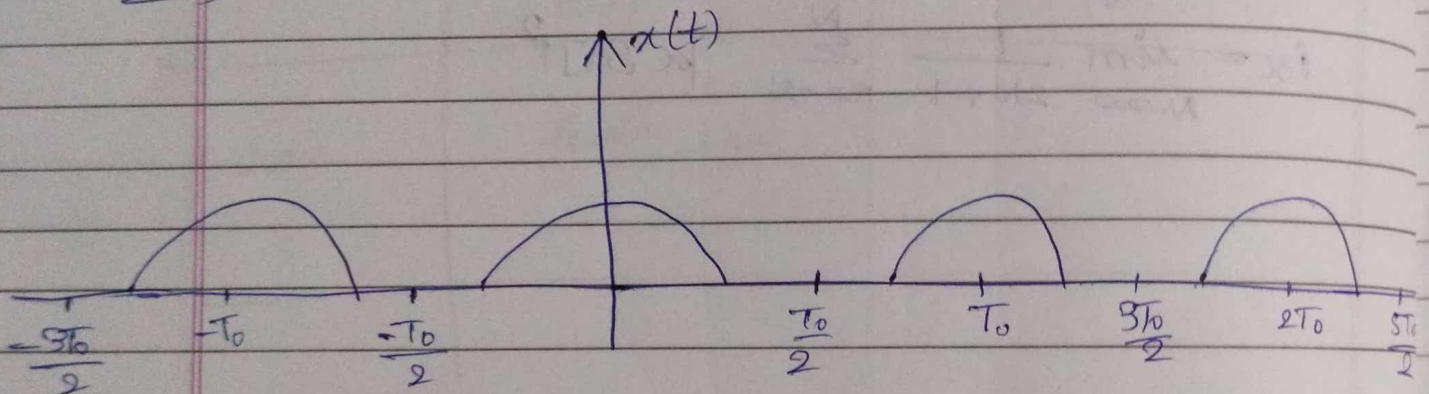
Ex-2) $x[n] = e^{j\omega_0 n}$

let $\frac{\omega_0}{2\pi} = p/q$

Sol $E_x = \lim_{N \rightarrow \infty} \sum_{n=-N}^N 1 = \lim$

Ex-3) Show that a periodic signal is always a power signal provided that energy per cycle is finite.

Sol $x(t) = x(t+T)$



$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{5T_0} \left[\int_{-3T_0/2}^{-T_0/2} |x(t)|^2 dt + \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt + \int_{T_0/2}^{3T_0/2} |x(t)|^2 dt + \dots \right]$$

(1) (2) (3)

$$= \lim_{n \rightarrow \infty} \frac{1}{nT_0} \left[\int_{-(n-2)T_0/2}^{(n-2)T_0/2} |x(t)|^2 dt + \int_{(n-2)T_0/2}^{nT_0/2} |x(t)|^2 dt \right]$$

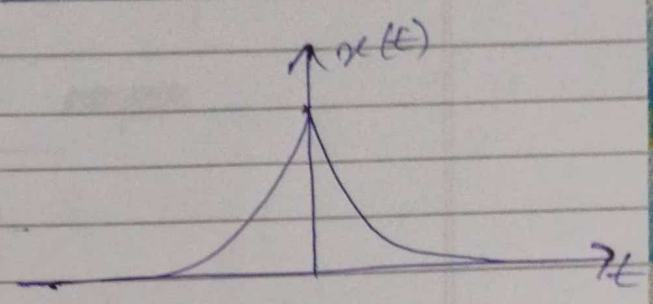
$$= \lim_{n \rightarrow \infty} \frac{1}{nT_0} nE = \lim_{n \rightarrow \infty} \frac{E}{T_0}$$

$$P_x = E/T_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt$$

* Power signals are periodic signals & energy signals are aperiodic signals.

Ex-4) $x(t) = e^{-|t|}$

Sol $\rightarrow E_x = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} e^{-2|t|} dt$



$$= \lim_{T \rightarrow \infty} 2 \int_0^{T/2} e^{-2t} dt = \lim_{T \rightarrow \infty} 2 \left[\frac{e^{-2t}}{-2} \right]_0^{T/2}$$

$$= \lim_{T \rightarrow \infty} 1 - e^{-T} = 1 = A_x^2$$

Ex-5) $x[n] = \begin{cases} n, & n > 0 \\ 0, & n \leq 0 \end{cases}$

Sol $\rightarrow E_x = \lim_{N \rightarrow \infty} \sum_{-N}^N |x[n]|^2$

$$= \lim_{N \rightarrow \infty} \sum_{n=0}^N n^2 = \infty$$

$$P_x = \frac{E_x}{2N+1} = \lim_{N \rightarrow \infty} \frac{\sum_{n=0}^N n^2}{2N+1} = \infty$$

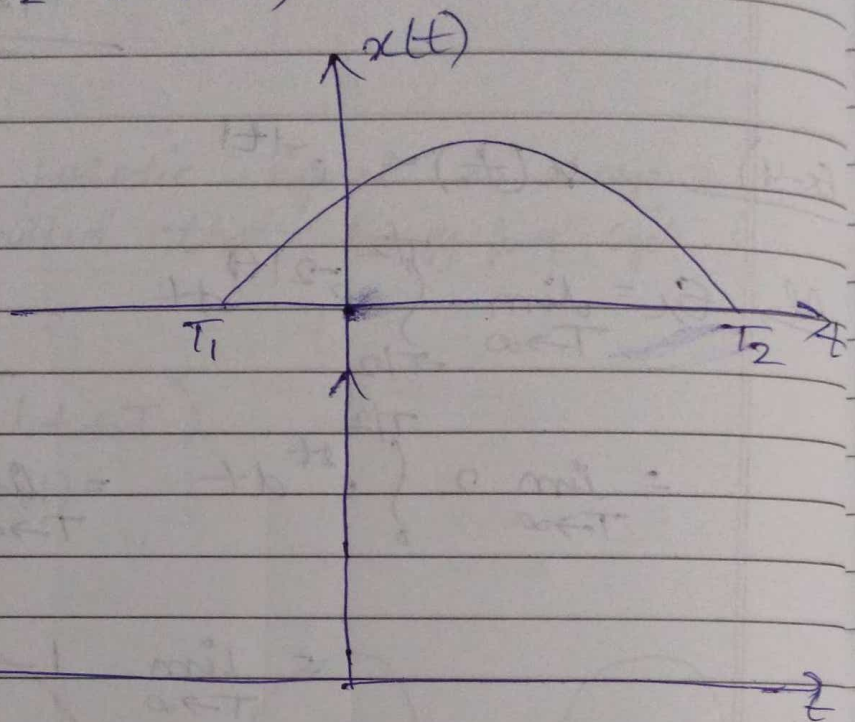
Neither energy nor power signal.

Transformation of independent variables:

① Shifting:

$$y(t) = x(t+b)$$

$$y(t) \Big|_{t=b} = x(0)$$



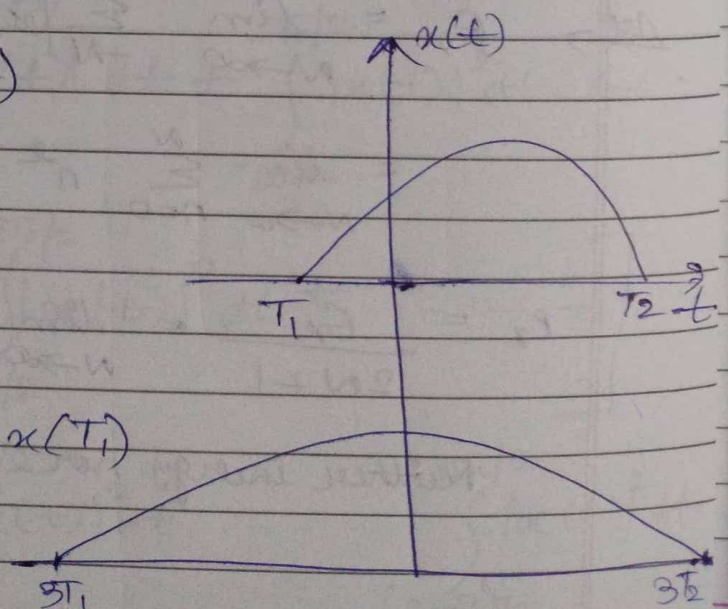
② Scaling:

$$y(t) = x(at)$$

$$\underline{0 < a < 1}$$

$$y(t) = x(t/3)$$

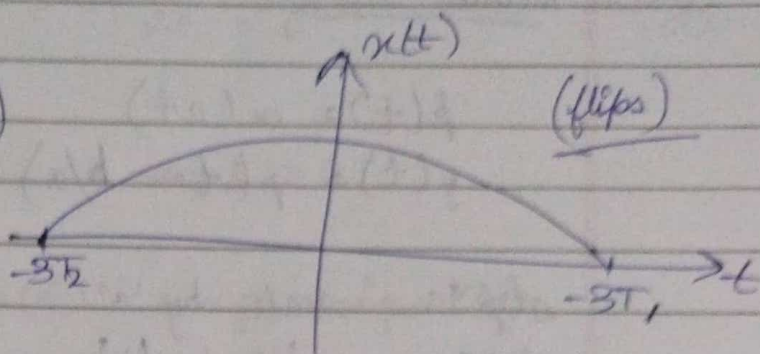
$$y(3T) = x\left(\frac{3T}{3}\right) = x(T)$$



$$\underline{-1 < a < 0}$$

$$y(t) = x(-t/3)$$

$$y(-3T_1) = x(T_1)$$



$|a| < 1 \rightarrow$ expansion ~~flip~~

$|a| > 1 \rightarrow$ compression

$a = -ve \rightarrow$ flip

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$$y(t) = x(t - b)$$

$$y(n) = x[n - n_0]$$

$$z(t) = x(at)$$

if $a = -ve \rightarrow$ flipping
 $a = +ve \rightarrow$ no flipping
 $|a| < 1 \rightarrow$ Expansion
 $|a| > 1 \rightarrow$ Compression

⑧ Time shifting & scaling:

$$y(t) = x(at + b) ; a > 0$$

(preferred method 1 -)

$$z(t) = x(t - b)$$

$$y(t) = z(at)$$

step 1: shift by 'b'.

step 2: scale by 'a'.

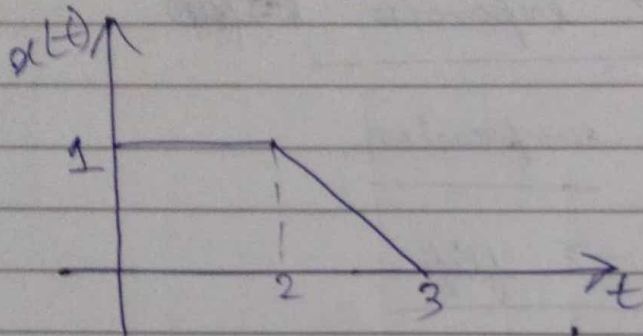
method 2 -

$$p(t) = x(at)$$

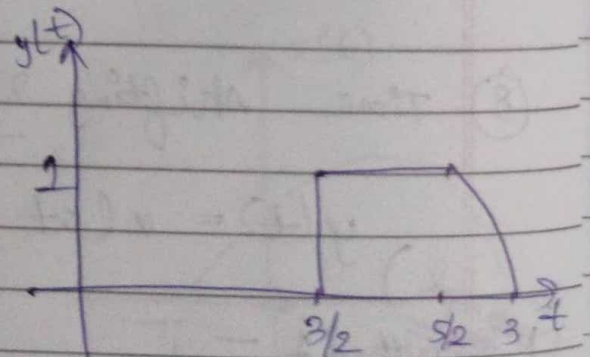
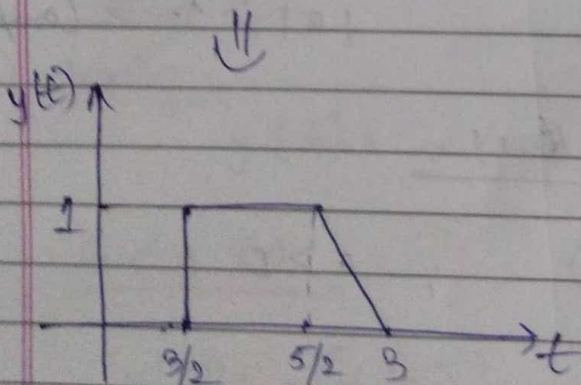
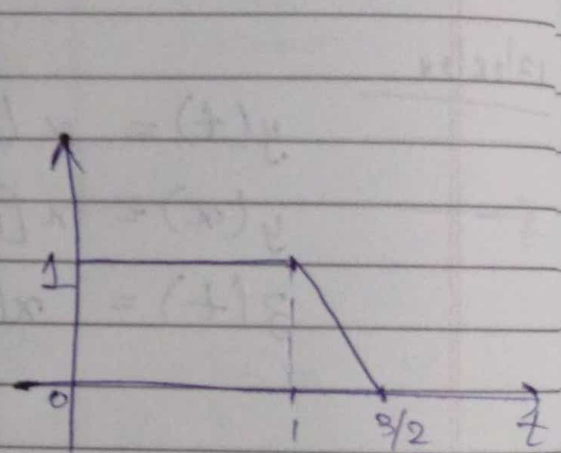
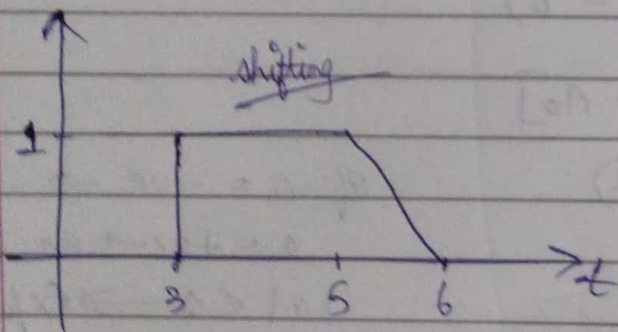
$$y(t) = p(t - b/a)$$

step 1: scale by 'a'

step 2: shift by $\frac{b}{a}$



$$y(t) = x(2t - 3)$$



$$y(3/2) = x(0)$$

$$y(2) = x(1)$$

Ex-1) Let $x(t)$ be an energy signal with energy E_x . Defining

(i) $y(t) = x(t - t_0)$

(ii) $z(t) = x(at - b)$ $a, b > 0$

(iii) $p(t) = x(-at)$

Are $y(t)$, $z(t)$, $p(t)$ energy signals?

sol (i) $E_y = \int_{-\infty}^{\infty} |y(t)|^2 dt = \int_{-\infty}^{\infty} |x(t - t_0)|^2 dt$ $t - t_0 = \tau$
 $= \int_{-\infty}^{\infty} |x(\tau)|^2 d\tau = E_x$

(ii) $E_z = \int_{-\infty}^{\infty} |x(at - b)|^2 dt$ $at - b = \tau$
 $= \int_{-\infty}^{\infty} |x(\tau)|^2 \frac{d\tau}{a}$ $dt = \frac{d\tau}{a}$
 $= \frac{E_x}{a}$

$y[n] = x[2n]$

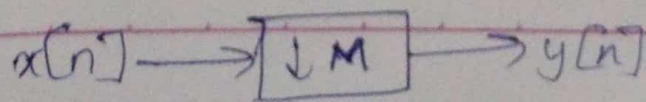
$x[n] = \{ \dots \underbrace{x_{-4} \ x_{-3} \ x_{-2} \ x_{-1} \ x_0 \ x_1 \ x_2 \ x_3}_{\uparrow} \dots \}$

$y[n] = \{ \dots \underbrace{x_{-8} \ x_{-6} \ x_{-4} \ x_{-2} \ x_0 \ x_2 \ x_4 \ x_6 \ x_8} \dots \}$

~~if~~ Continuous time process is a loss-less process whereas discrete time process is a ~~loss~~ losing process.

Decimation / down sampling (in case of compression)

$y[n] = x[Mn]$

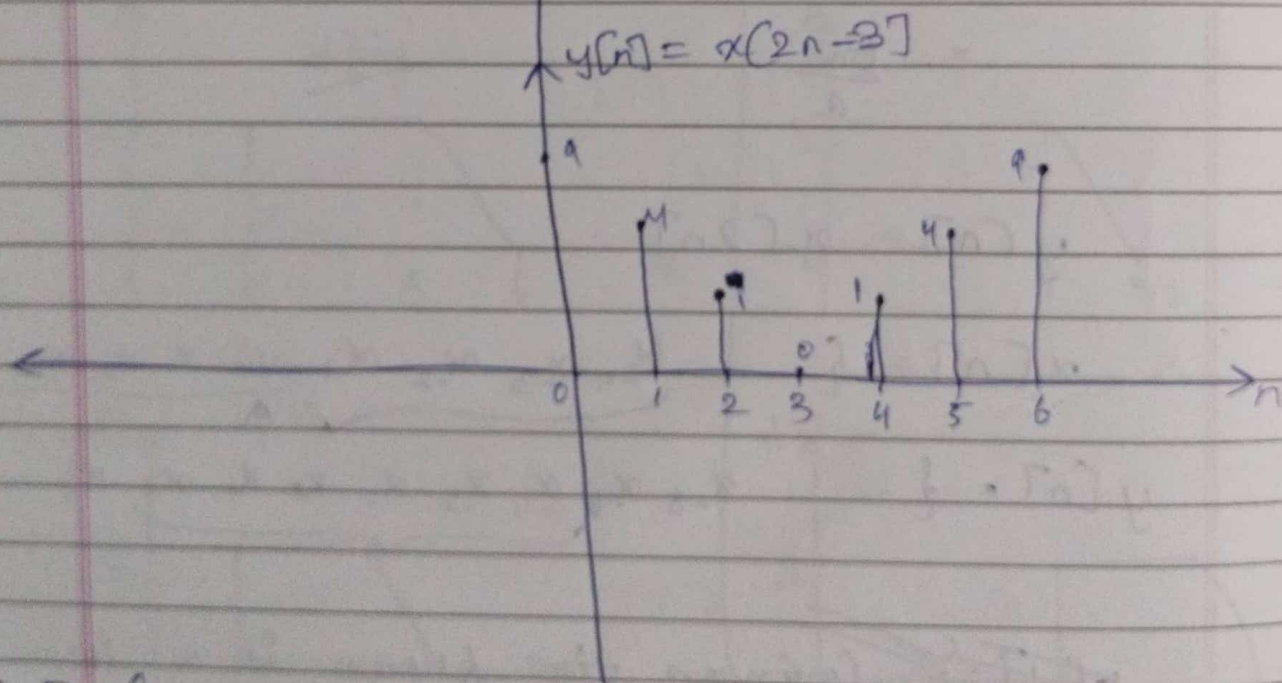
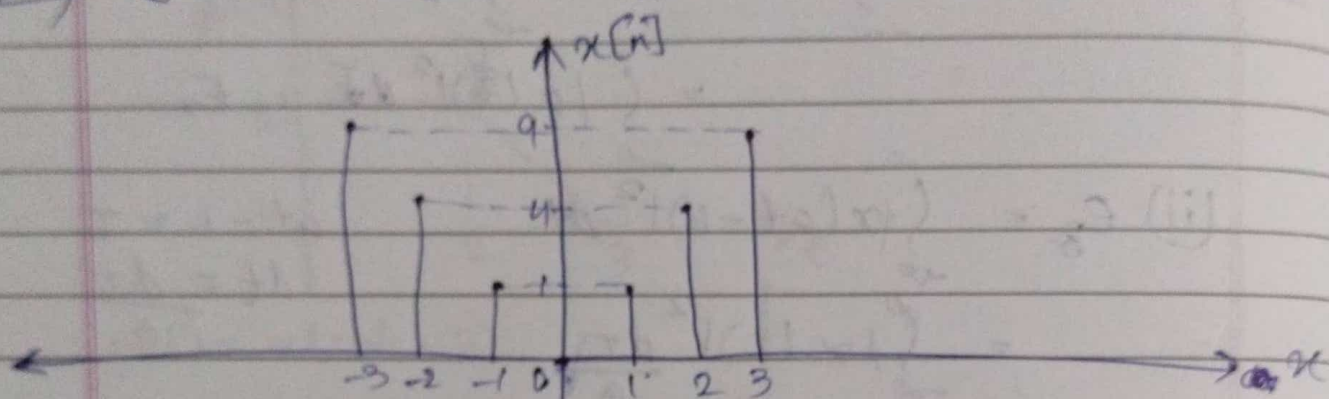


Interpolation or up sampling: (in case of expansion)

$$p[n] = \begin{cases} x[n/2] & ; n \text{ even} \\ 0 & ; n \text{ odd} \end{cases}$$

$$x[n] \rightarrow \boxed{\uparrow M} \rightarrow p[n]$$

Q) $y[n] = x[2n-3]$ $x[n] = n^2$



$$x[n] = \{ \dots, 9, 4, 1, 0, 1, 4, 9, 16, 25, \dots \}$$

$$x[n-2] = \{ \dots, 0, 9, 0, 4, 0, 1, 0, 0, 0, 1, 0, 4, 0, 9, 0, 16, 0, \dots \}$$

Vectors:

$$\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k} \quad \text{or} \quad \vec{v} = [v_x \ v_y \ v_z]^T$$

Any arbitrary vector

$$x(t) = \sum_{k=-\infty}^{\infty} x_k \phi_k(t) \quad \text{--- orthogonal series expansion}$$

1. $\phi_k(t)$ is also periodic.
2. $\phi_k(t)$ should have some physical significance.
3. $\langle \phi_k(t), \phi_j(t) \rangle \rightarrow 0$ Inner product
4. $\langle \phi_k(t), \phi_k(t) \rangle \rightarrow 1$

INNER PRODUCT (dot product of vectors):

a) for continuous time periodic signals:

$$\frac{1}{T} \int_{-T/2}^{T/2} x(t) y^*(t) dt$$

$$\Rightarrow \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = P_x = 1$$

b) for discrete time periodic signal:

$$\frac{1}{N} \sum_{n=0}^{N-1} x[n] y^*[n]$$

c) for CT aperiodic signal:

$$\int_{-\infty}^{\infty} x(t) y^*(t) dt$$

d) for DT aperiodic signal:

$$\sum_{n=-\infty}^{\infty} x[n] y^*[n]$$

Complex exponential function:

1) CT complex exponential function:

$$\phi(t) = e^{j\omega_0 t}$$

- ① (i) Always periodic / periodic for all values of ω_0
 (ii) Power signal with unit power.
 ② (iii) As $\omega_0 \uparrow$, $T_0 \downarrow$ so, rate of change of signal increases, so frequency increases.

$$\left| \frac{d}{dt} \phi(t) \right| = \left| j\omega_0 e^{j\omega_0 t} \right| = \omega_0$$

As ω_0 increases from 0 to ∞ , its frequency also increases from 0 to ∞ . So, frequency $\propto \omega_0$.

(iv) All ω_0 values are distinct from 0 to ∞ .

③ Harmonically related:

A set of complex exponentials are said to be harmonically related if they are periodic over a common time period.

$$\begin{cases} e^{j\frac{2\pi}{T_0}t} \Rightarrow T_0 \\ e^{j\frac{2\pi p}{T_0}t} \Rightarrow T_0/p \end{cases} \therefore \text{also periodic w.r.t } T_0$$

$$\boxed{e^{j\frac{2\pi}{mT_0}t} \Rightarrow mT_0}$$

④	$e^{jk\omega_0 t}$	$e^{jp\omega_0 t}$	$e^{j\omega_0 t}$
	$\left(\begin{smallmatrix} k\text{th} \\ \text{harmonic} \end{smallmatrix} \right)$	$\left(\begin{smallmatrix} p\text{th} \\ \text{harmonic} \end{smallmatrix} \right)$	$\left(\begin{smallmatrix} \text{fundamental} \\ \text{harmonic} \end{smallmatrix} \right)$

TP: $\frac{T_0}{k}$

$\frac{T_0}{p}$

T_0

$$\frac{1}{T} \int_{-T/2}^{T/2} x(t) y^*(t) dt$$

(T = continuous time period)

$$\langle e^{jk\omega t}, e^{jp\omega t} \rangle = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} e^{j(k-p)\omega t} dt = 0$$

$$= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} (\cos m\omega t + j \sin m\omega t) dt$$

$$= 0$$

$$\langle e^{jk\omega t}, e^{jp\omega t} \rangle = \delta[k-p]$$

$$x(t) = \sum_{k=-\infty}^{\infty} x_k \phi_k(t) ; \phi_k(t) = e^{jk\omega t}$$

$$x_k = 0 \text{ for } |k| > 200 \rightarrow \text{Low pass signal}$$

$$x_k = 0 \text{ for } |k| < 200 \rightarrow \text{High pass signal}$$

2) DT complex exponential function:

$e^{jn\omega_0}$

① (i) Not always periodic.

$$\text{Periodic iff } \frac{2\pi}{\omega_0} = \frac{n}{p} \Rightarrow \omega_0 = \frac{2\pi}{N}$$

$$\text{② } \text{Period} = 1$$

$$\text{③ } \omega_1 = \omega_0 + 2\pi$$

$$e^{jn\omega_1} = e^{jn\omega_0} \cdot e^{jn2\pi} = e^{jn\omega_0}$$

Result:

All ro's are not distinct and DT frequencies are distinct only in the range of 2π .

(4) Rate of change of $e^{j\frac{2\pi}{N}n}$

$$\begin{aligned}
 & e^{j\frac{2\pi}{N}n} - (e^{j\frac{2\pi}{N}n} \cdot e^{-j\frac{2\pi}{N}}) \\
 &= e^{j\frac{2\pi}{N}n} \left\{ 1 - e^{-j\frac{2\pi}{N}} \right\} = e^{j\frac{2\pi}{N}n} \left\{ e^{j\frac{\pi}{N}} - e^{-j\frac{\pi}{N}} \right\} e^{-j\frac{\pi}{N}} \\
 &= e^{j\frac{2\pi}{N} \left(n - \frac{\pi}{2} \right)} \cdot 2j \sin \left(\frac{2\pi}{2N} \right)
 \end{aligned}$$

(Rate of change)

$$= \left| 2 \sin \left(\frac{2\pi}{2N} \right) \right|$$

→ Rate of change would be max^m when $\frac{2\pi}{2N} = \pi/2$

$$\Rightarrow \boxed{\frac{2\pi}{N} = \pi}$$

→ Rate of change would be min^m when

$$\frac{2\pi}{2N} = 0, \pi$$

$$\Rightarrow \boxed{\frac{2\pi}{N} = 0, 2\pi}$$

So, frequency ~~is not~~ ^{increases} from 0 to π & decreases from π to 2π .

(5) Harmonics — functions that are periodic over the same time period are called harmonics.

$$e^{j\frac{2\pi}{N}n} \Rightarrow TP = N$$

$$\phi_k[n] = e^{j\frac{2\pi}{N}kn} \Rightarrow \phi_k[n+m] = e^{j\frac{2\pi}{N}kn} \cdot e^{j\frac{2\pi}{N}km} = e^{j\frac{2\pi}{N}k(n+m)} = \phi_k[n+m]$$

$$\Rightarrow e^{j\frac{2\pi}{N}km} = 1$$

$$\phi[n] = e^{j \frac{2\pi}{N} n}$$

fundamental harmonic

$$\phi_k[n] = e^{j \left(\frac{2\pi k}{N} \right) n}$$

kth harmonic

$$\begin{aligned} \phi_{k+N}[n] &= e^{j \left(\frac{2\pi}{N} (k+N) n \right)} \cdot \cancel{e^{j \frac{2\pi}{N} n}}^1 \\ &= \phi_k[n] \end{aligned}$$

$$x[n] = \sum_{k=0}^{N-1} X_k \phi_k[n]$$

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \phi_k(t)$$

⑥ Orthogonality property -

$$\frac{1}{N} \sum_{n=0}^{N-1} x[n] y^*[n]$$

$$\frac{1}{N} \sum_{n=0}^{N-1} e^{j \frac{2\pi}{N} kn} \cdot e^{-j \frac{2\pi}{N} pn}$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} e^{j \frac{2\pi}{N} (k-p)n}$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} e^{j \frac{2\pi}{N} mn}$$

$$\frac{1}{N} \sum_{n=0}^{N-1} (a)^n \quad a = e^{j \frac{2\pi m}{N}}$$

$$\frac{1}{N} \left\{ \frac{a^N - 1}{a - 1} \right\}$$

$$a^N = e^{j2\pi m} = 1$$

$$x[n] = \sum_k x_k e^{j\frac{2\pi}{N}nk}$$

$$= \sum_k |x_k| e^{j(\frac{2\pi}{N}nk + \phi_k)}$$

$$a = |a| e^{j\phi}$$

$$= a_k + j b_k$$

~~1/2~~

$$\sin \omega_0 t = \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t}$$

$$\frac{1}{2j} = -\frac{1}{2}j = \frac{1}{2} e^{j3\pi/2}$$

Exercises

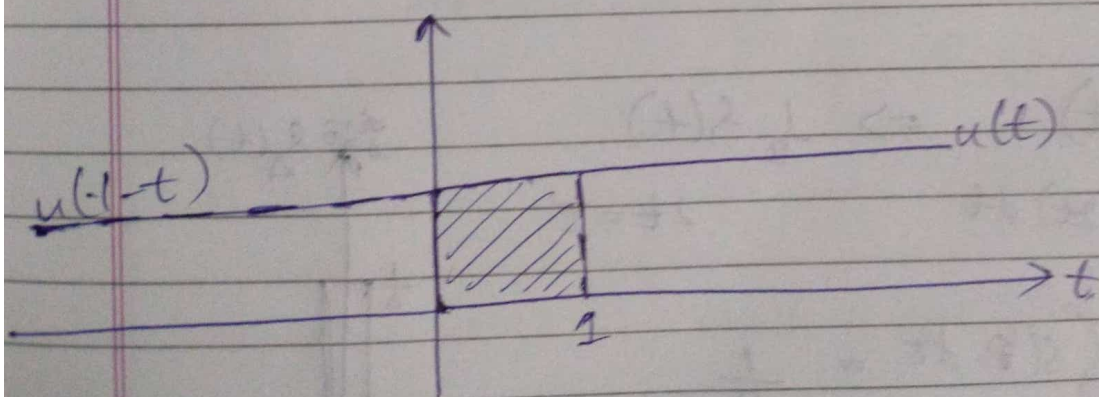
1) sketch the signal $y(t) = \sum_{n=0}^{\infty} u(t-2n)u(1+2nt)$

$$z(u) = \sum_{x=-\infty}^{\infty} x(t-xT_0)$$

Sol →

$$z(t) = u(t)u(1-t)$$

$$\sum_{n=0}^{\infty} z(t-2n) = \sum_{n=0}^{\infty} u(t-2n)u(1-t+2n)$$



2) $x[n]$ is periodic with $TP = N$,
 $y[n]$ is defined as $y[n] = (-1)^n x[n]$
 Is $y[n]$ periodic??

3) $x(t) = \cos\{\sin(4\pi t)u(t)\}$
 Is $x(t)$ periodic?

4) Evaluate:

(i) $\int_{-\infty}^{\infty} \delta(t-2) \delta(x-t) dt$ (ii) $\int_{-3}^6 (4-t)^2 \delta(t+4) dt$

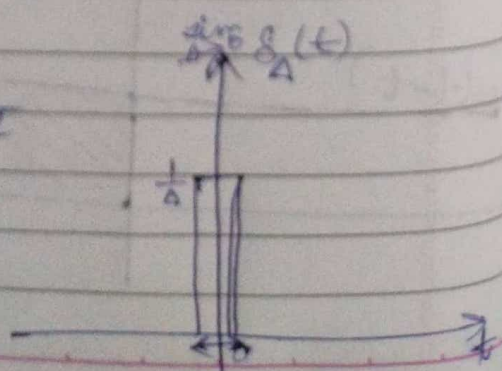
5) Represent $\delta(2t)$ in terms of $\delta(t)$.

6) Represent $\delta[2n]$ in terms of $\delta[n]$.

Answers

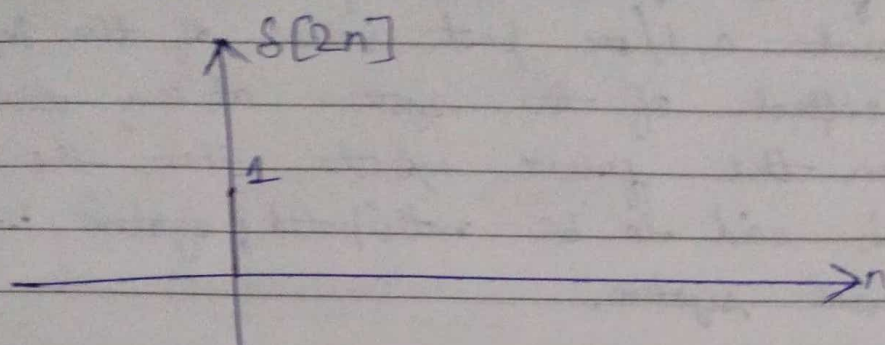
4)(ii) $\int_{-3}^6 (4-t)^2 \delta(t+4) dt$
 $= \int_{-3}^6 (4+4)^2 \delta(t+4) dt$; $t+4 = 0$
 $\Rightarrow t = -4$

5) $\delta(2t) \Rightarrow \frac{1}{2} \delta(t)$
 $\int_{-\infty}^{\infty} \delta(2t) dt$ $2t = \tau$
 $= \frac{1}{2} \int_{-\infty}^{\infty} \delta(\frac{\tau}{2}) d\tau = \frac{1}{2}$



6) $\delta(2t)$

$$\delta[2n] = \delta[n]$$



ex/08/24

~~no~~ $y[n] = \frac{1}{2n_0+1} \sum_{k=n-n_0}^{n+n_0} x[k] \Rightarrow$ Moving average

sol $x_1[n] \rightarrow \frac{1}{2n_0+1} \sum_{k=n-n_0}^{n+n_0} x_1[k]$

$x_2[n] \rightarrow$ " " $x_2[k]$

$x_3[n] \rightarrow$ " " $x_3[k]$

a) Linearity let $x_3[n] = ax_1[n] + bx_2[n]$
 $y_3[n] = ay_1[n] + by_2[n] \Rightarrow$ LINEAR

b) TI let $x_3[n] = x_1[n-M]$

$$y_3[n] = \frac{1}{2n_0+1} \sum_{k=n-n_0}^{n+n_0} x_1[k-M] = \frac{1}{2n_0+1} \sum_{p=n-M-n_0}^{n-M+n_0} x_1[p]$$

Property - 3)

Causality: A system is said to be a causal if the output at any time depends on the present and/or past values of the input. If the output of the system at any time depends on the future inputs, then the system is said to be anticipated system, i.e., a non-causal system.

$$y(t) / y[n]$$

$$x(t) / x[n]$$

$$y(t) = \sum_{k=-\infty}^0 x[k] \rightarrow \text{causal system}$$

$$y(t) = \sum_{k=0}^{\infty} x[-k] \rightarrow \text{causal system}$$

Ex: 1) $y[n] = \frac{1}{2n_0+1} \sum_{k=n-n_0}^{n+n_0} x[k]$

$$= \frac{1}{2n_0+1} \left\{ \underbrace{x[n-n_0]}_{\text{Past}} + \underbrace{x[n]}_{\text{Present}} + \underbrace{x[n+n_0]}_{\text{Future}} \right\}$$

So, non-causal system

Ex: 2) $y(t) = \int_{-\infty}^t x(2\tau-3) d\tau$
 $= \int_{-\infty}^t 2x(4p-3) dp$

Let $\tau = 2p$

Case 1: $t=0$, $y(0) = \int_{-\infty}^0 2x(4p-3) dp \Rightarrow x(-\infty) - x(-3)$

Case 2: $t < 0$, $y(-2) = \int_{-\infty}^{-2} 2x(4p-3) dp \Rightarrow x(-\infty) - x(-1)$

Case 3: $t > 0$, $y(2) = \int_{-\infty}^2 2x(4p-3) dp \Rightarrow x(-\infty) - x(5)$

So, non-causal system

Ex-3) $y(t) = u x(t)$

$$= \begin{cases} 1 & x(t) > 0 \\ 1/2 & x(t) = 0 \\ 0 & x(t) < 0 \end{cases}$$

So, causal system

Property-1) Stability - A system is said to be BIBO stable (bounded input bounded output) if the output is bounded for all values of t or n (time) given that the input is bounded.

$$|y[n]| < \infty \quad \text{if} \quad |x[n]| < \infty \quad \forall n$$

~~$|y[n]|$~~

$$|y(t)| < \infty \quad \text{if} \quad |x(t)| < \infty \quad \forall t$$

Ex- $y[n] = \frac{1}{2n_0+1} \sum_{k=n-n_0}^{n+n_0} x[k]$

Sol $\rightarrow$ Bounded input $\Rightarrow |x[n]| \leq B_x \quad \forall n$

$$|y[n]| = \frac{1}{2n_0+1} \left| \sum_{k=n-n_0}^{n+n_0} x[k] \right| \leq \frac{1}{2n_0+1} \sum_{k=n-n_0}^{n+n_0} |x[k]|$$

$$\leq \frac{1}{2n_0+1} \sum_{k=n-n_0}^{n+n_0} B_x = B_x$$

$$\Rightarrow |y[n]| \leq B_x$$

So, stable system.

Ex-2) $y(t) = \int_{-\infty}^{2t} x(2\tau - 3) d\tau$

$$= \int_{-\infty}^t 2x(4p - 3) dp$$

Sol $|y(t)| \leq \int_{-\infty}^t 2|x(4p - 3)| dp$

$$|x(t)| \leq B_x$$

$$|x(2t)| \leq B_x$$

$$\leq \int_{-\infty}^t 2B_x dp$$

$$= 2B_x [p]_{-\infty}^t$$

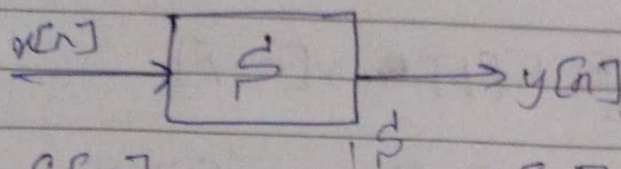
$$= 2B_x [t + \infty] = \infty$$

So, unstable system

Ex-3) $y(t) = u\{x(t)\}$

$$= \begin{cases} 1, & x(t) > 0 \\ 1/2, & x(t) = 0 \\ 0, & x(t) < 0 \end{cases}$$

So, stable system

~~TFI~~LTI systems

- $s[n]$ | $h[n]$
 $a s[n]$ | $a h[n]$ linearity
- $s[n-k]$ | $h[n-k]$ TI (time invariance)
- $x[n] = \sum_{k=-\infty}^{\infty} x[k] s[n-k]$ Convolution property
- $x[k] s[n-k]$ | $x[k] h[n-k]$ Homogeneity + TI
- $x[p] s[n-p] + x[k] s[n-k]$

- $$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$
Convolution sum

- $$y(t) = x(t) * h(t)$$

$$= \int_{-\infty}^{\infty} x(t) h(t-\tau) d\tau$$
Convolution integral

→ $h(t)/h[n]$ completely characterises / represents the CT/DT LTI system.

$$\begin{aligned}
 y[n] &= x[n] * h[n] \\
 &= \sum_k x[k] h[n-k] \quad n=k=p \\
 &= \sum_p h[p] x[n-p] \\
 &= h[0] x[n] + x[n]
 \end{aligned}$$

$$y(t) = \int x(\tau) h(t-\tau) d\tau$$

Ex-1) Compute $u(t) * u(t)$

Sol → Method 1: Analytical

$$\begin{aligned}
 u(t) * u(t) &= \int_{-\infty}^{\infty} u(\tau) u(t-\tau) d\tau \quad \begin{matrix} u(t) = 0 \\ \tau < 0 \end{matrix} \\
 &= \int_0^{\infty} u(t-\tau) d\tau \quad \begin{matrix} u(t-\tau) = 0, t-\tau < 0 \\ \tau > t \end{matrix} \\
 &= 1, \tau < t
 \end{aligned}$$

$$= \int_0^t 1 d\tau, t > 0$$

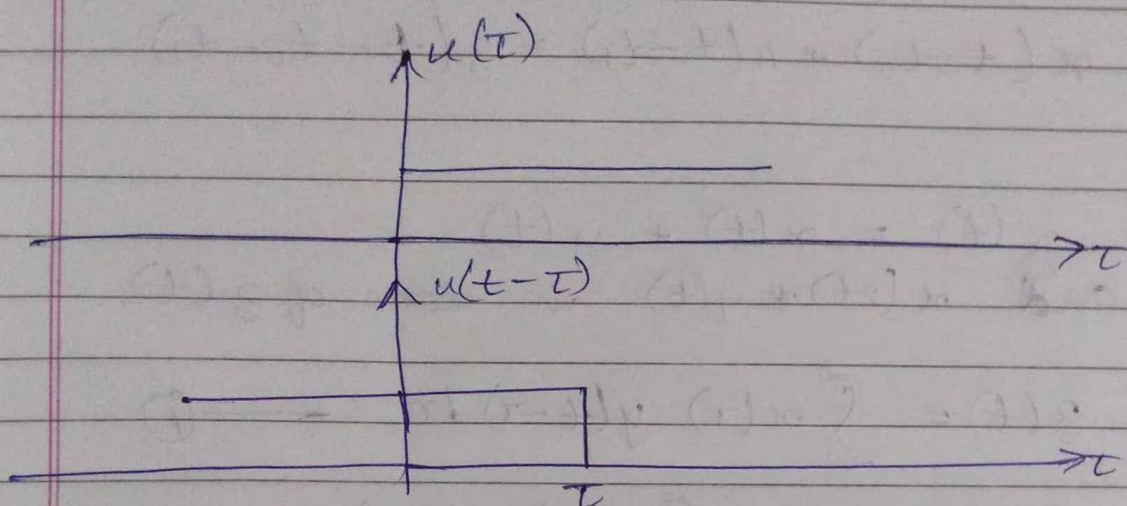
$$= t, t > 0$$

$$= tu(t)$$

Method 2: graphical

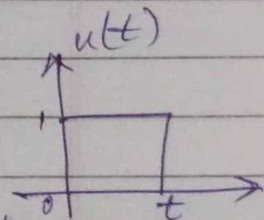
$$y(t) = u(t) * u(t)$$

$$= \int_{-\infty}^{\infty} u(\tau) u(t-\tau) d\tau$$



Case 1: $t < 0$, $y(t) = 0$

Case 2: $t > 0$, $y(t) = \int_0^t 1 d\tau = t$



Results

$$\rightarrow \delta(t) * u(t) = u(t)$$

$$\rightarrow u(t) * u(t) = tu(t)$$

$$\rightarrow \delta[n] * u[n] = u[n]$$

$$\rightarrow u[n] * u[n] = nu[n]$$

Properties

① $x(t) * y(t) = y(t) * x(t)$

② $x(t) * y(t) = z(t)$

then, $x(t-t_0) * y(t-t_1) = z(t-t_0-t_1)$

Let $z(t) = x(t) * h(t)$

Compute $x(t-t_0) * h(t-t_1)$ in terms of $z(t)$

Sol $\rightarrow z(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$ ——— ①

$$x(t-t_0) * h(t-t_1) = \int_{-\infty}^{\infty} x(\tau-t_0) h(t-\tau-t_1) d\tau$$

put $\tau - t_0 = \alpha$

$$= \int \alpha(\alpha) h(t - t_0 - t_1 - \alpha) d\alpha \quad \text{--- (2)}$$

Comparing (1) & (2) we get,

$$x(t - t_0) * h(t - t_1) = z(t - t_0 - t_1)$$

Ex) $z(t) = x(t) * y(t)$
find $x(2t) * y(t)$ in terms of $z(t)$.

Soln $z(t) = \int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau \quad \text{--- (1)}$

$$x(2t) * y(t) = \int_{-\infty}^{\infty} x(2\tau) y(t - \tau) d\tau \quad 2\tau = \alpha$$

$$= \int_{-\infty}^{\infty} x(\alpha) y(t - \alpha/2) d\alpha \quad \text{--- (2)}$$

$$x(2t) * y(2t) =$$